

Stock Price Prediction using Deep Learning and Sentiment Analysis

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Abstract—Shares of publicly traded businesses are traded on the stock market. An accurate prediction of stock price movement produces profits for investors. Due to being dynamic and nonlinear in nature, accurate stock price prediction is very challenging and complex. However, studies in this field demonstrate that the stock price may be forecast with some degree of accuracy. This paper presents a stock price prediction model using deep learning and news sentiment analysis. Firstly, the Long Short-Term Memory (LSTM) architecture was applied on NIFTY 50 data imported from Yahoo finance (pertaining to March 2017 - March 2022) to obtain the batch/epoch settings that ensured the best performance. Secondly, news sentiment data was imported for the same period and analysed for its polarity and subjectivity. Thirdly, CAT Booster and Random Forests were applied on the finance dataset augmented with news sentiment parameters to predict stock values during March 2022 to April 2023 based on the last closing prices of March 2022. Primary results indicate that the LSTM model with 100 epochs and 6 layers yielded the best prediction accuracy based on past financial trends whereas the CAT Booster model performed marginally better than Random Forest in predicting stock values based on past financial data and news sentiments.

Index Terms—Cat booster, random forest, deep learning, LSTM, NIFTY, stock price prediction, sentiment analysis

I. INTRODUCTION

The advent of the fourth industrial revolution has resulted in large scale automation enabled by artificial intelligence and machine learning in various fields [1]–[4]. These tools are being successfully applied to financial sector as well. The

stock market refers to various exchanges where shares of listed companies are bought and sold. Stock markets are an essential part of a free-market economy because they allow democratized access to investor transactions and the exchange of capital. Stock market forecasts aim to determine the future development of the value of shares in the financial markets. Predicting stock market movements is one of the most challenging problems because forecasting stocks involves many factors, such as interest rates, politics, and economic growth, identify applicable funding agencies here which can make the stock market volatile and challenging to predict with precision. However, accurate stock market predictions can bring profits to investors. In order to create more precise and trustworthy stock market forecasting techniques, researchers analyse financial data and news sentiment using machine learning and deep learning models including recurrent neural networks (RNNs), decision trees, gradient boosting, and LSTM in combination with neuro-linguistic programming (NLP) methods. Machine learning and deep learning algorithms have been successfully applied in a variety of domains [5]–[9]. The novelty of this approach is the combination of deep learning and NLP, which enables the extraction of insightful information from textual data, allowing for a more thorough comprehension of market dynamics and sentiments.

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train = train.drop(['Adj Close'], axis=1)
train.head()

```

	Date	Open	High	Low	Close	Volume
0	2017-03-15	9086.648809	9106.549805	9075.500000	9084.799805	240800
1	2017-03-16	9129.650391	9158.460195	9128.549805	9153.700195	208000
2	2017-03-17	9207.799805	9218.400391	9147.599809	9160.049805	286700
3	2017-03-20	9166.950195	9167.599609	9116.299805	9126.849609	332100
4	2017-03-21	9133.950195	9147.750000	9087.200195	9121.500000	247700

```

print("Null values:", train.isnull().values.sum())
print("NA values:", train.isna().values.any())

Null values: 0
NA values: False

train.info()

<class 'pandas.core.frame.DataFrame'>
RangeIndex: 1232 entries, 0 to 1231
Data columns (total 6 columns):
#   Column   Non-Null Count  Dtype
---  ---
0   Date     1232 non-null   datetime64[ns]
1   Open     1232 non-null   float64
2   High     1232 non-null   float64
3   Low      1232 non-null   float64
4   Close    1232 non-null   float64
5   Volume   1232 non-null   int64
dtypes: datetime64[ns](1), float64(4), int64(1)
memory usage: 57.9 KB

```

Fig. 1. Training dataset (Yahoo finance)

II. LITERATURE REVIEW

Kumar et al. (2019) analysed the performance of various deep learning models, including LSTM, GRU and hybrid models in predicting Nifty 50 stock values in order to find the best effective deep learning technique [10]. Akita and Yang (2018) conducted a thorough study on the use of deep learning methods such as RNNs, CNNs and DBNs for stock price prediction, evaluating the techniques' success in various stock markets such as the Nifty 50 and outlining the field's future directions. Mishra et al. (2021) and others examined how well different deep learning architectures such as LSTM, CNN, and attention-based models predicted the prices of the Nifty 50 stock index [11]–[13]. The interpretability and generalizability of these models were examined in the context of Nifty 50 stock price prediction as the authors assessed them using various performance indicators. Thapar and Kumar (2020) [14] concentrated on LSTM networks for forecasting Nifty 50 stock values, contrasting them with conventional time-series forecasting methods and examining the influence of various input variables on prediction accuracy. The study demonstrated how well LSTM models predict the price of the Nifty 50 stock. Singh and Gupta (2022) [15] proposed a hybrid deep learning strategy for Nifty 50 stock market prediction that integrated LSTM, CNN, and SVR. Hybrid model demonstrated increased prediction accuracy for Nifty 50 stock values when compared to separate deep learning models and conventional machine learning techniques [16]. A recent study by Moghar et al. [17]) showed that the RNN-based LSTM is very promising as the model is capable of predicting prices very effectively. Another investigation by Mehtab et al. [18]) showed that the

satisfactory prediction of stock values using LSTM regression model on NIFTY 50 data. According to Mehtab et al., stock predictions made using LSTM and sentiment analysis had an accuracy range of 85 to 95 percent [19]. Many studies also measured prediction accuracies using a mean absolute error (MAE) values [20], [21]. The combination of deep learning with news sentiment analysis gave a boost in prediction accuracy. Hence, in the present study, LSTM, CAT Booster and Random Forest models were applied on financial data along with news sentiment parameters and prediction accuracies were expressed in terms of root mean square error (RMSE).

III. METHODOLOGY

This section presents the detailed step wise methodology followed in the current study.

A. Preparation of data set

Firstly, financial data was imported from yahoo finance for 2017 to 2022 tenure. After import, the dataset's head and tail were suitably indexed followed by data cleaning that included removing the not required variables like 'adj close' and null values. In this study, the NIFTY 50 data from 13 March 2017 to 13 March 2022 was extracted from Yahoo Finance and was cleaned as described above (Figure 1). The variables derived in the dataset after cleaning were Open Price, Close Price, High Price and Low Price. Similarly, the testing dataset was prepared of the same NIFTY 50 from 4 March 2022 to 14 April 2023.

Comparison Between Stock High and Low Price

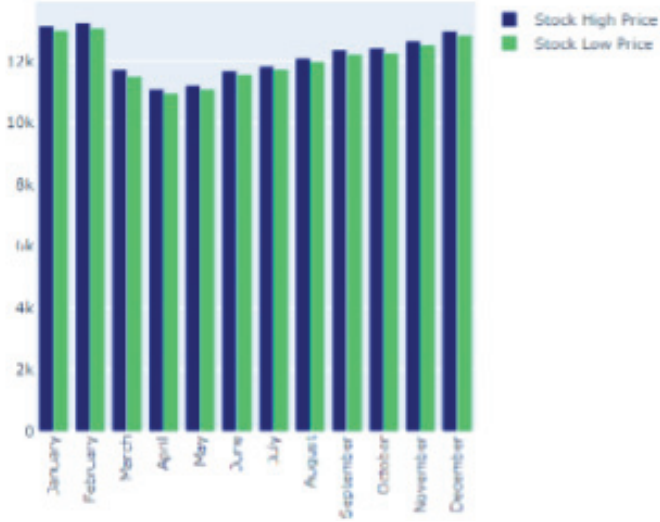


Fig. 2. High and low prices (Yahoo finance)

Comparison Between Stock Open and Close price

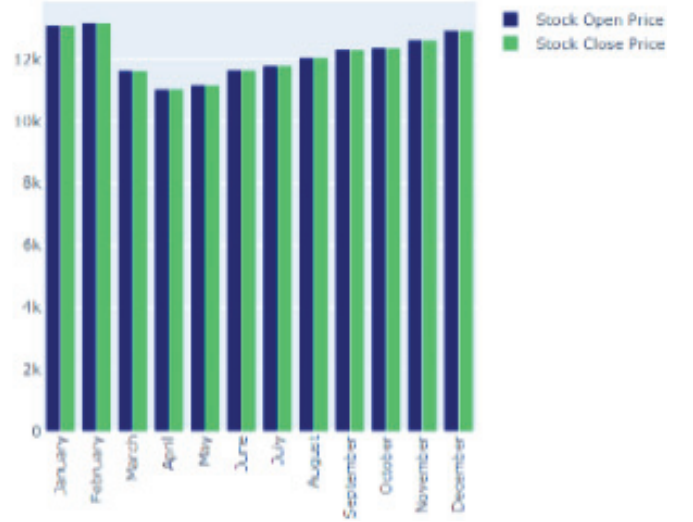


Fig. 3. Open and close prices (Yahoo finance)

B. Data Visualization

Data visualisation highlights the underlying patterns and trends in the dataset using charts, plots, heat maps, and moving averages [22], [23]. These visualisations are made possible by programs such as Matplotlib, Seaborn, Plotly, Tableau and Google Data Studio [24]–[26]. In this study, month-wise averages of the open and close stock price variables are shown in figure 2, whereas a comparison of high price and low price variables are shown in figure 3. These figures show minor changes in the average open and close prices as well as among the high and low prices across months. Figure 3 shows the overall NIFTY 50 index during the last five years. The plot shows steady growth over the years, except for a temporary dip during the Covid period.

C. Prediction

This study employed recurrent neural networks (RNN) for NIFTY 50 stock price predictions. These neural networks are effective at representing sequence data, including time series. Each piece of information is retained over time by the RNN. The ability to recall past input makes it helpful for time series predictions. The LSTM model is a type of RNN architecture. This model provided the projected price graph, the mean absolute error of forecast and the actual testing data set. Mean absolute error (MAE) is a metric that measures the overall prediction accuracy by quantifying the average absolute difference between projected and actual values. To measure public opinion and its possible effect on market movements and investor behaviour, news sentiment analysis of the stock market entails analysing and assessing the emotional tone or sentiment exhibited in news pieces directly connected to the stock market. Hence, in this study news sentiment data was sorted according to the NIFTY companies using a web crawler

which provided NIFTY-impacted news. This data was used to define the news-based polarity and subjectivity of each NIFTY stock, and the subsequent results were added to the base stock price data to be implemented in further models (i) Random Forest (ii) CAT Booster. The prediction accuracies of these models were evaluated using the root mean squared error, i.e. the average squared difference between anticipated and actual values. This metric provides a measurement of the overall prediction accuracy by giving greater weight to bigger mistakes.

IV. LSTM MODEL

The Long Short-Term Memory (LSTM) artificial neural networks consist of feedback loops in contrast to the conventional feedforward neural networks. LSTM networks can effectively analyse complete data sequences, not just individual data points. Because of this feature, LSTM networks are excellent candidates for sequential data processing and prediction. For instance, LSTMs may be used for remedial therapy, gaming video, sound admission robot control, content recognition, voice recognition, machine translation, and classification. In the present work, the Sklearn and MinMaxScaler were imported into LSTM for the initial training of the financial dataset with a training range of (60, 1232). Reshaping of the dataset is a very major part of training. LSTM was applied by reshaping the dataset with a set of 50 units with a dropout unit of 0.2. This process created a proper shape of data for max predictions and was obtained through trial-and-error. Firstly, different kinds of combinations of the LSTM model with different epochs, batch sizes and layers of neural networks were tried. Different combinations like 100 epochs, batch size 32 and 8 layers; 100 epochs, batch size 32 and 6 layers; 100 epochs, batch size 64 and 4 layers; 150 epochs, batch size 32 and 6 layers were tried. The four-layered LSTM model

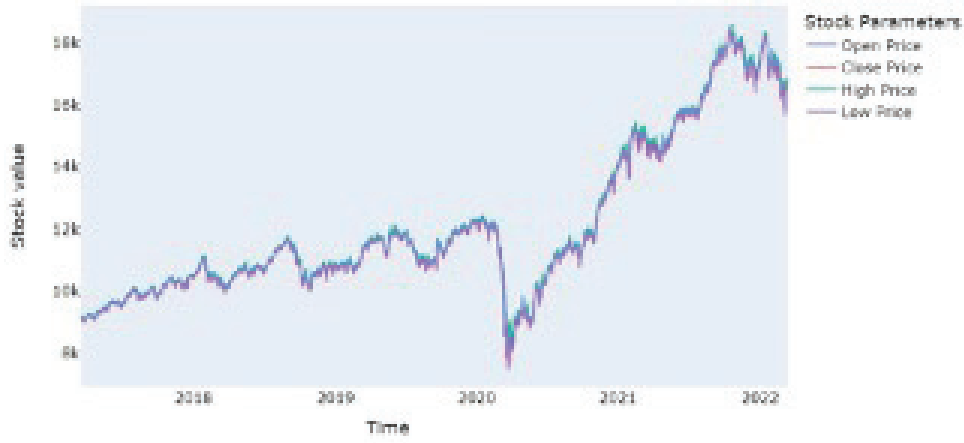


Fig. 4. NIFTY movement (Yahoo finance)

with 100 epochs and batch size of 64 was found to yield the minimum mean absolute error of 0.018. The dropout option was used because it provides the feasibility to drop out that amount of data if there is over-fitting in running the program. The data was finally trained with a shape of (1172,60), since this shaping resulted in the perfect LSTM model for further predictions. The adam optimizer was applied for obtaining the minimum mean squared error and the financial data was fitted into a batch size of 64 and 100 epochs. The description of the model is given in Figure 5. A similar visualization and cleaning process was carried out for the testing data set as well, which ranged from 4-3-2022 to 11-4-2023. The test period stock prices were predicted based on the training data set and the predicted prices were compared to the actual prices of those dates and finally, the prediction performance was obtained with mean 0.883, standard deviation 0.077 and MAE 0.017.

V. NEWS SENTIMENTS ANALYSIS

In the second phase, sentimental analysis for stock price prediction was carried out by analysing newspaper headlines in the finance sector during the specified period. The CSV file of news sentiments from March 2017 to March 2022 along with news data parameters was loaded from the web crawler. This data was used to define the positive or negative sentiment of a particular stock as well as its confidence level. The sentiments data file was also cleaned of all null values so that only NIFTY-related sentiments were present in it. The cleansed news sentiments dataset was merged with the financial reshaped training dataset for polarity and subjectivity analyses. Polarity in sentimental analysis relates to the emotion that is conveyed in a text and can be either positive, negative, or neutral. Typically, polarity is expressed as a numerical value between -1 (most negative) and 1 (most positive), with 0 denoting a neutral feeling. It technically depends on what the sentiments say and the graph of the polarity of the dataset considered in the present study indicated its neutrality as shown in figure 6. Subjectivity, on the other hand, refers to the extent to

```
lstm_model.summary()
```

Model: "sequential"

Layer (type)	Output Shape	Param #
lstm (LSTM)	(None, 60, 50)	10400
dropout (Dropout)	(None, 60, 50)	0
lstm_1 (LSTM)	(None, 60, 50)	20200
dropout_1 (Dropout)	(None, 60, 50)	0
lstm_2 (LSTM)	(None, 60, 50)	20200
dropout_2 (Dropout)	(None, 60, 50)	0
lstm_3 (LSTM)	(None, 50)	20200
dropout_3 (Dropout)	(None, 50)	0
dense (Dense)	(None, 1)	51

Total params: 71,051
Trainable params: 71,051
Non-trainable params: 0

Fig. 5. LSTM summary

which a piece of text is opinionated or expresses a personal point of view. It is typically represented by a score ranging from 0 (most objective) to 1 (most subjective). It is given by examining proper prices as well as opinions. The subjectivity graph of the dataset considered in the present study was high (in the range of 0 to 1) and negligible towards the negative side which indicated a rise in prices and hence the corresponding positive sentiments are reflected in figure 7. On the basis of subjectivity and polarity analyses, the final dataset file was prepared which included the subjectivity, polarity, negative, positive and neutral values of the sentiments of all stocks (figure ??). The closing prices were specified in the dataset which formed the basis of future price predictions. This sentiment cum financial stock price dataset was compiled for the final predictions using two other models i.e. Cat Booster and Random Forest.

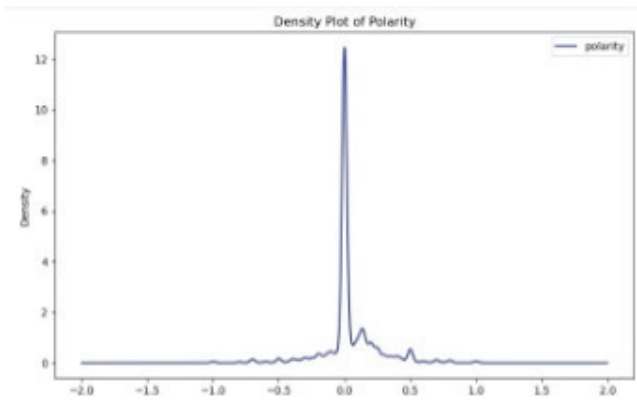


Fig. 6. Polarity

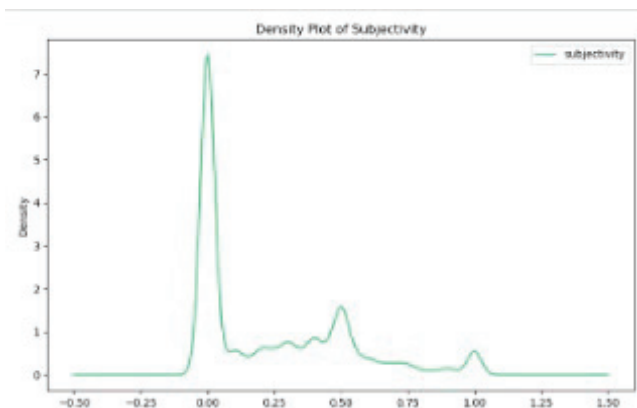


Fig. 7. Subjectivity

VI. CAT BOOSTER AND RANDOM FOREST MODELS

Cat boosting is based on the approach of decision trees and gradient boosting. The main idea behind this scheme is to create a competitive model that helps the search request by combining it with a few weaker models that execute better than singular models in terms of classification. Cat Boost can be used for sorting, guidance forecasts, and as personal assistants as well. In the present study, the cat booster was successfully implemented in code and was applied on to the sentiment-financial dataset which resulted in an RMSE values of 0.154. The random forest method obtains the best result by selecting the largest decision tree with the best performance values. Random forests (RFs) are often used to predict the accuracy/classification of remote sensing data. In the present study, random forest was successfully implemented in the code and was applied on to the sentiment-financial dataset which resulted in an RMSE values of 0.158.

VII. DISCUSSIONS

In this study, an LSTM model was applied on a base NIFTY 50 stock price data followed by merging its dataset with the news sentiments dataset obtained from a web crawler, that resulted in a new dataset which included both the stock prices as well as the sentimental polarities of that day. This dataset was applied in CAT Booster and Random Forest models

which resulted in their respective prediction RMSE values mentioned in the previous section. These values are quite low and demonstrate the successful stock price prediction abilities of the proposed methodology which combines financial stock price dataset with news sentiments for deep learning. The lowest prediction error values were obtained using the CatBoost model and hence this model is recommended for future stock price predictions. This research investigated the use of various deep learning approaches for stock price prediction, including LSTM-RNN, Cat Booster and Random Forest. The proposed methodology delivered accurate stock price predictions by combining news sentiment analysis and utilizing statistical and deep learning methodologies. This methodology might lead to better investment decisions, stock market forecasting and potentially bigger earnings for investors. This study emphasises the necessity of considering public opinion and its potential impact on market movements and investor behaviour by including news sentiment analysis into the stock price prediction models. Analysing and interpreting the emotional tone or attitude represented in stock market news provides useful insights for projecting stock values. It addresses incorporating macro and micro aspects, such as global economic conditions, unforeseen events, political crises, and company financial performance with sentiment research to improve stock price prediction accuracy. It enables the deep learning models to capture the complexity and dynamics of the stock market by taking into account a wide range of parameters. This study evaluated various models such as LSTM, Cat Booster, and Random Forest using mean absolute error and root mean squared error. The importance of model evaluation in stock price prediction research is shown by the selection of the model with the lowest error for accurate forecasts. Constructing a backend stock price prediction software is a major implication of this study as these models can be used as a program that drives an application or website or an interface that forecasts the future close price of overall NIFTY on the basis of the provided sentiments and span of data and suggests future investment decisions. The usefulness of such studies hinges on the availability and calibre of the data, which includes sentiment labelling and news items. It might be difficult to get a trustworthy dataset for sentiment analysis, but taking latency and real-time data into account is crucial. It is also challenging to capture and quantify all external elements such as unanticipated occurrences, geopolitics, market mood and legislations that affect stock values. The practicality and predictive power of the models can be ensured by including and accounting for these elements. The present study primarily used metrics such as MAE and RMSE to assess model performances. While these measures show how accurate a forecast is, they cannot accurately reflect the model's usefulness and profitability in actual trading circumstances. A more thorough examination would result from the inclusion of other evaluation criteria, such as risk-adjusted returns or profitability measurements. Moreover, deep learning models are prone to overfitting, especially when working with sparse data or when the complexity of the model is not well handled.

More investigation is needed to discuss how overfitting affects the model's performance and investigate ways to alleviate this problem, such as regularisation, cross-validation, or ensemble approaches.

VIII. CONCLUSIONS AND FUTURE SCOPE

This study presented a stock price prediction methodology based on deep learning and news sentiment analysis. Stock price data pertaining to the period March 2017 - March 2022 was used for predicting the prices during March 2022 to April 2023 period. Yahoo finance and a web crawler were used to source the stock price and new sentiment datasets respectively. Firstly, the datasets were cleaned of null values and irrelevant variables. The cleaned data was visualized using monthly averages to indicate minor stock price deviations and steady growth over the years. Thereafter, various combinations of epochs, batch sizes and layers were tried in LSTM architectures to obtain the best prediction performance. The four layered LSTM with batch size 64 and 100 epochs emerged as the best performing architecture with a mean of 0.883, standard deviation 0.077 and MAE 0.017. In the next step, news sentiment analysis was carried out. Firstly, the sentiment data was cleaned of null values. After proper visualization of the data, subjectivity and polarity of the sentiments were determined which resulted in the negative, positive and neutral values of the NIFTY stocks. The resultant stock price cum sentiment parameter dataset was applied in the cat booster and random forest models for forecasting. The root means square error value was 0.015 in CAT booster regression and 0.016 in random forest. Hence, the minimum root means square error value was given by random forest regression. The future scope of this research is sectorial analysis and sector-wise price predictions. For this purpose, the NIFTY dataset can be divided into various sectors such as banking, technical, foods and beverages etc. and determine for which model is predicts the best for which sector. The news sentiments can also be analysed in a similar sector-wise manner by using the web crawler. Another future scope of this research is to make a front-end website app for this program that takes an input of a particular date's price, and predicts a future price for NIFTY applying for this program in the back end. This would give the user a proper predicted price for a future date and can prove to be a perfect application of this research.

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